

Chapter 02: Sensor Fundamentals

Robotics III: Sensors and Perception in Robotics

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Characteristics of Sensors

Range
Sensitivity
Repeatability
Response
Resolution
Accuracy
Bandwidth
Linearity

Classification of Sensors

Active	passive	
Analog	digital	
Proprioceptive	Exteroceptive	
Optical	Magnetic	Mechanical

Outline

Sensor Fundamentals

- Sensor Definition
- Sensors in Closed-Loop Feedback Control
- Characteristics of Sensors
- Analog-to-Digital Conversion
- Classification of Sensors
- Sensors and Perception in Robotics: Task and Challenges
- Examples

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Sensors – Definition

- Sensor: (lat.: Sensus = „capable of sensitivity“)
- A sensor is a device that can detect/sense and **measure physical properties** from the environment such as temperature, luminance, weight, distance, velocity, force, size, ...
- Sensors deliver **low-level information** about the robot's environment. This information is limited, inaccurate and noisy
- Therefore, sensors return a low-level **incomplete description** of the world

Sensor – Definition

- Sensors are physical devices that
 - receive a **signal or stimulus** and
 - react to it with an **electrical signal**
- Any sensor is an **energy converter**
 - No matter what you try to measure, you always deal with energy transfer from the object of measurement to the sensor
- Simple and complex sensors, in terms of information they provide:
 - A switch is a simple on/off sensor
 - A human retina is complex sensor consisting of more than a hundred million photosensitive elements (rods and cones)

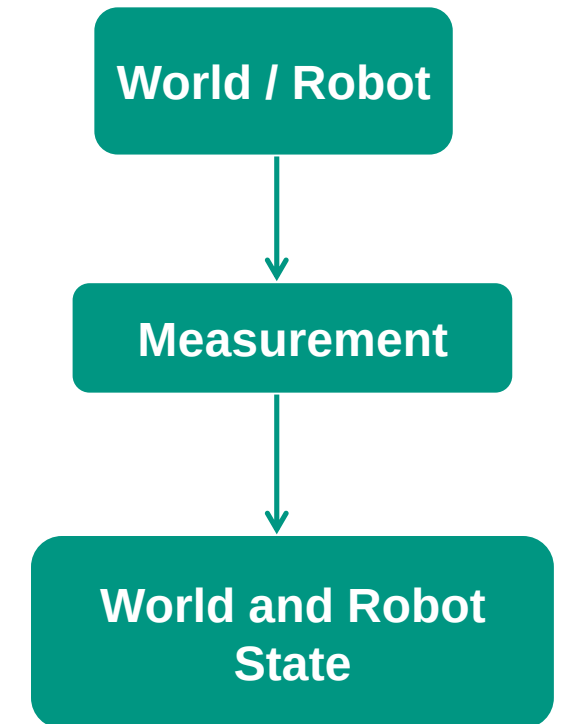
Sensor – Definition

- Sensors constitute the perception system of a robot
- Sensors are fundamental for closed-loop feedback control
- A **robot's intelligence** depends on
 - the quality, diversity and richness of the information provided by its sensors
 - the ability to process the sensor data efficiently (in real-time)
- Types of senses are called **sensory modalities**
 - Multi-modal perception integrates complementary sensor data for enhanced environmental understanding and robustness

Sensors – Definition

From raw physical measurements and high-level semantic understanding

- Sensors capture the attributes of the world
- Sensors provide raw data which must be transformed into semantic information to determine the state of the environment or the robot
- Signal processing is needed
 - Feature extracting to identify relevant patterns
 - Pattern recognitions to classify detected feature
 - Filtering to reduce noise and enhance signal quality
 - Data fusion to integrate multiple sensor inputs
 - ...



Outline

Sensor Fundamentals

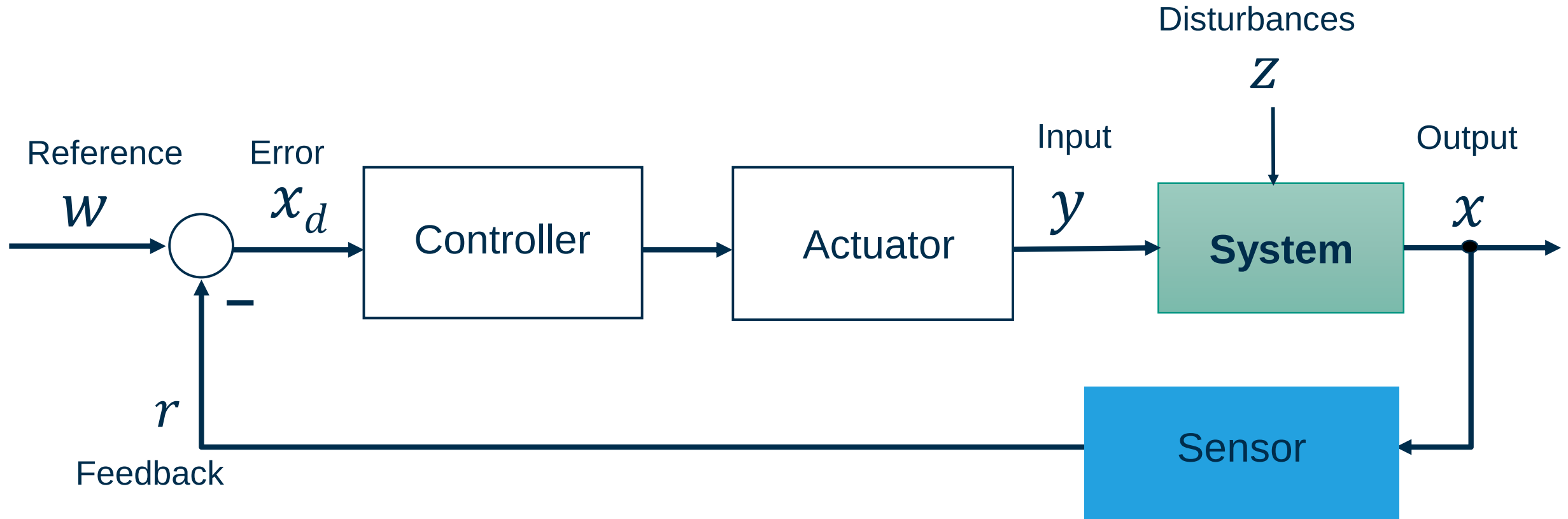
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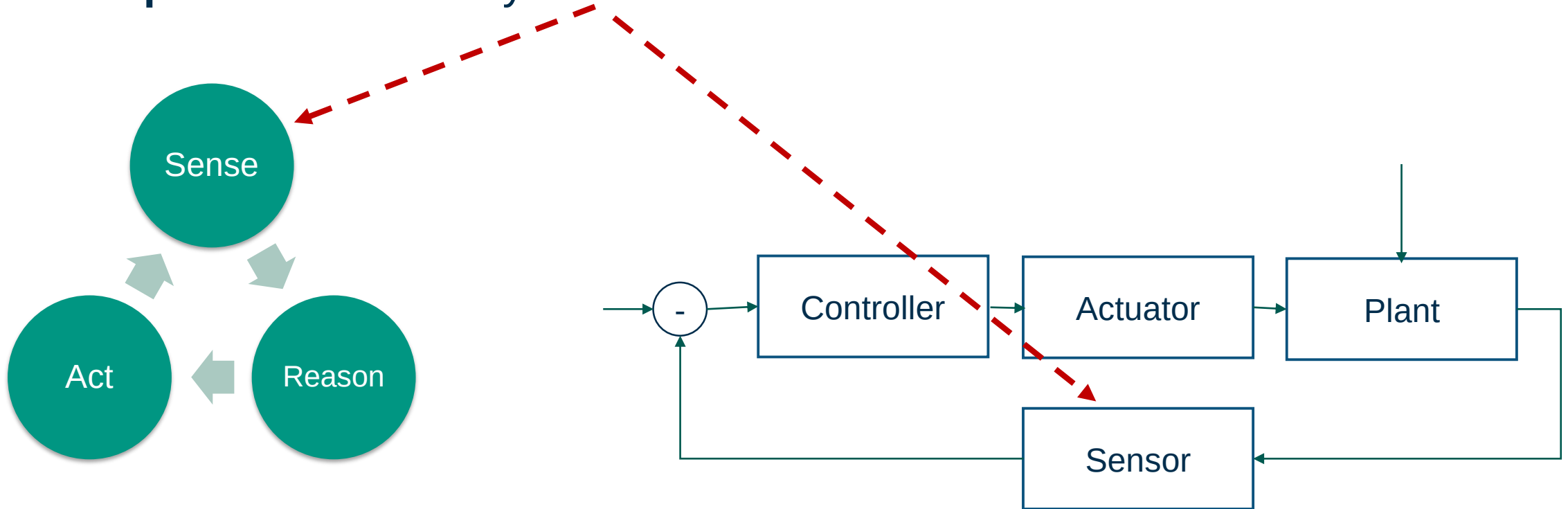
Sensors in Closed-Loop Feedback Control



Without sensors, the system (robot) would operate "blindly" in open-loop control
→ unable to detect errors or adapt to disturbances
→ impossible operation in real-world, dynamic environments

Sensors in Robotics

- **Closed-Loop Control** plays a fundamental role in robotics
- **Sensors** are always part of a closed-control loop
- **Perception** is enabled by **sensors**



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Characteristics of Sensors

- Range
- Resolution
- Sensitivity
- Accuracy
- Repeatability
- Bandwidth
- Response
- Linearity

Range/Span

Range of input signals that can be measured and converted

Range = Maximum measurable value – Minimum measurable value

Sensor Type	Physical Property	Typical Range	Example Application
LIDAR	Distance	0.1 m – 100 m	Obstacle detection & SLAM
IMU (Gyroscope)	Angular velocity	$\pm 250^\circ/\text{s}$ to $\pm 2000^\circ/\text{s}$	Balance in legged robots
Force Sensor	Force	0 – 200 N	Force feedback in manipulation tasks
Ultrasonic Sensor	Distance	2 cm – 400 cm	Short-range navigation

Resolution

- Smallest change in the input signal that can be detected. High resolution allows more precise measurements

$$\text{Resolution} = \frac{\text{Range}}{\text{Number of quantization levels}}$$

- **Example:** An incremental joint encoder with 1024 pulses per revolution (10 Bit) has a resolution of

$$\frac{1 \text{ revolution}}{1024 \text{ pulses}} \times \frac{360 \text{ degrees}}{\text{revolution}} = 0,3516 \frac{\text{degrees}}{\text{pulse}}$$

- Resolution can be specified in two ways:
 - **Absolute:** see above
 - **Relative:** divided by the maximum measurable value (measuring range end)

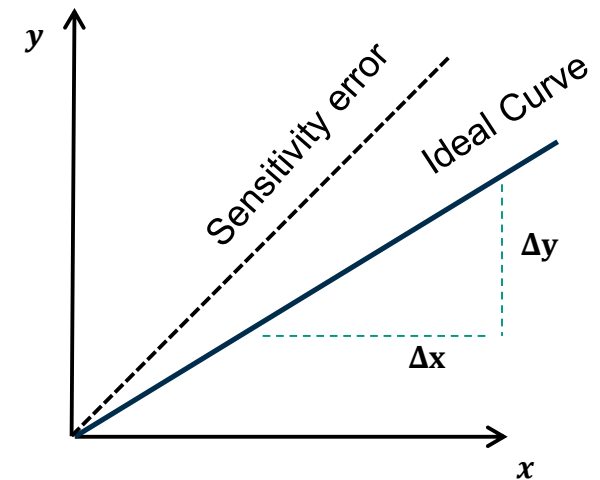
$$\text{Resolution}_{\text{rel}} = \frac{\text{Range}}{\text{Number of quantization levels}} \times \frac{1}{\text{Maximum measurable value}}$$

Sensitivity

- Change of the sensor output relative to a change in the input signal

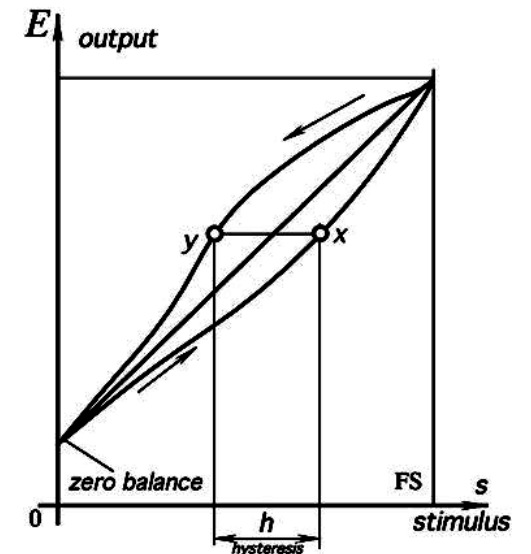
$$\text{Sensitivity} = \frac{\Delta \text{Output}}{\Delta \text{Input}}$$

- A linear sensor has constant sensitivity over the entire range
- Sensitivity is given through the characteristic curve of a sensor
 - Example: A temperature sensor with 10 mV change per 1°C has the sensitivity of 10 mV/°C
- **Sensitivity error:** Deviation from the ideal sensitivity curve
 - Example: the temperature sensor above have an actual sensitivity of 11.2 mV/°C instead of 10 mV/°C → Error = 1.2mV/°C



Accuracy

- Deviation between measured value and true value of the measured quantity
- Accuracy quantifies how close sensor readings are to the actual physical reality
- Error sources can be:
 - **Bias (Offset):** Constant error over the entire measurement range
 - **Hysteresis (Hysterese):** Output depends on the direction of change
 - **Random noise:** unpredictable variations in the sensor output
- Accuracy can be given either as a percentage of full scale or in absolute terms



Hysteresis

From Fraden, J.: Handbook of Modern Sensors

Repeatability

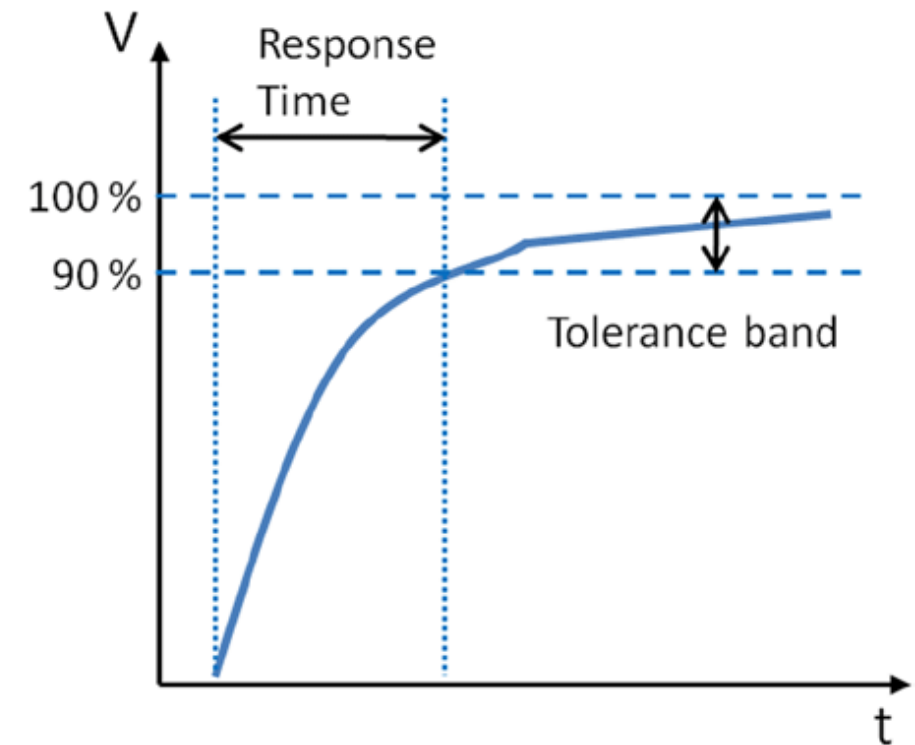
- The ability of a sensor to reproduce the same output when the same input conditions are applied repeatedly over short period, under the same operating conditions
- Repeatability is often expressed as the **standard deviation or range of repeated measurements** for a fixed input. Lower variability indicates better repeatability.

Bandwidth

- Refers to the range of frequencies over which the sensor can detect and respond to input signals
- Bandwidth represents the sensor's ability to **capture dynamic changes in the measured quantity**
- Defined as difference between the lowest and highest frequency, i.e. slowest and fastest change in the input signal that can be correctly measured by the sensor.
- **Examples**
 - Accelerometers: From <1 Hz to >10 kHz (application dependent)
 - Temperature sensors: Typically 0.1-10 Hz (limited by thermal mass)
 - Pressure sensors: From <1 Hz to several kHz
 - Optical sensors: Can range from Hz to GHz depending on technology

Response Time

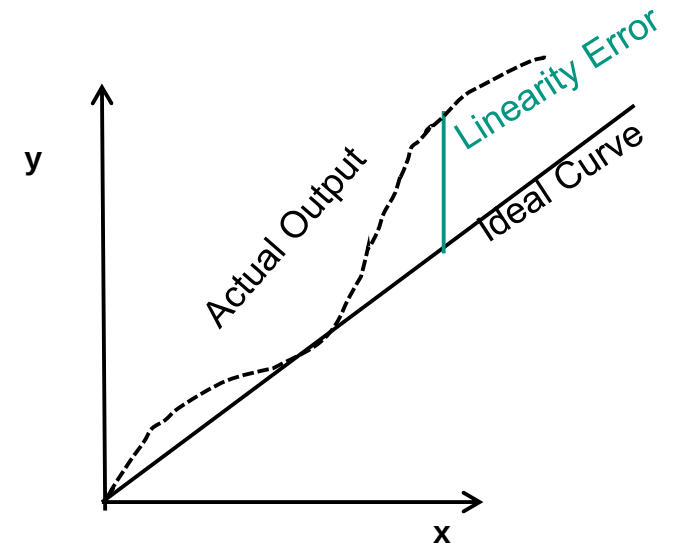
- Time delay (T_r) from change in input to change in output
- Response time is the time interval required for a sensor's output to change from its initial state to a specified percentage (typically 90% or 95%, within a tolerance band) of its final steady-state value after a step change in the input quantity being measured.



McGrath, Michael J., et al. "Sensing and sensor fundamentals." Sensor technologies: Healthcare, wellness, and environmental applications (2013): 15-50.

Linearity

- Linearity is the measure of how closely a sensor's output-input dependency follows a linear relationship across the operating range. It quantifies the **maximum deviation** between the sensor's actual response curve and an ideal linear response.
- **Ideal response:** $Output = k \cdot Input$
- A perfect linear sensor would maintain a constant ratio between input change and output change throughout its entire range
- Deviation measurement: Quantifies how much the actual response deviates from the ideal linear response



Outline

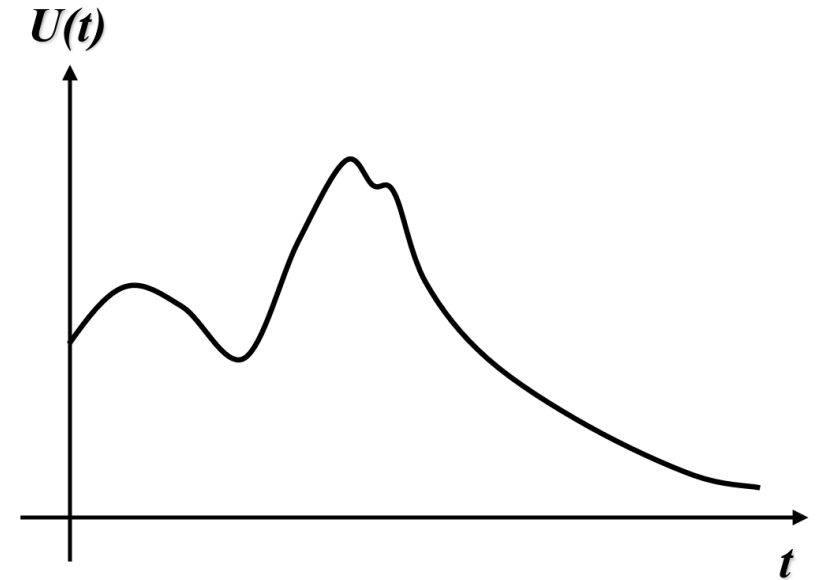
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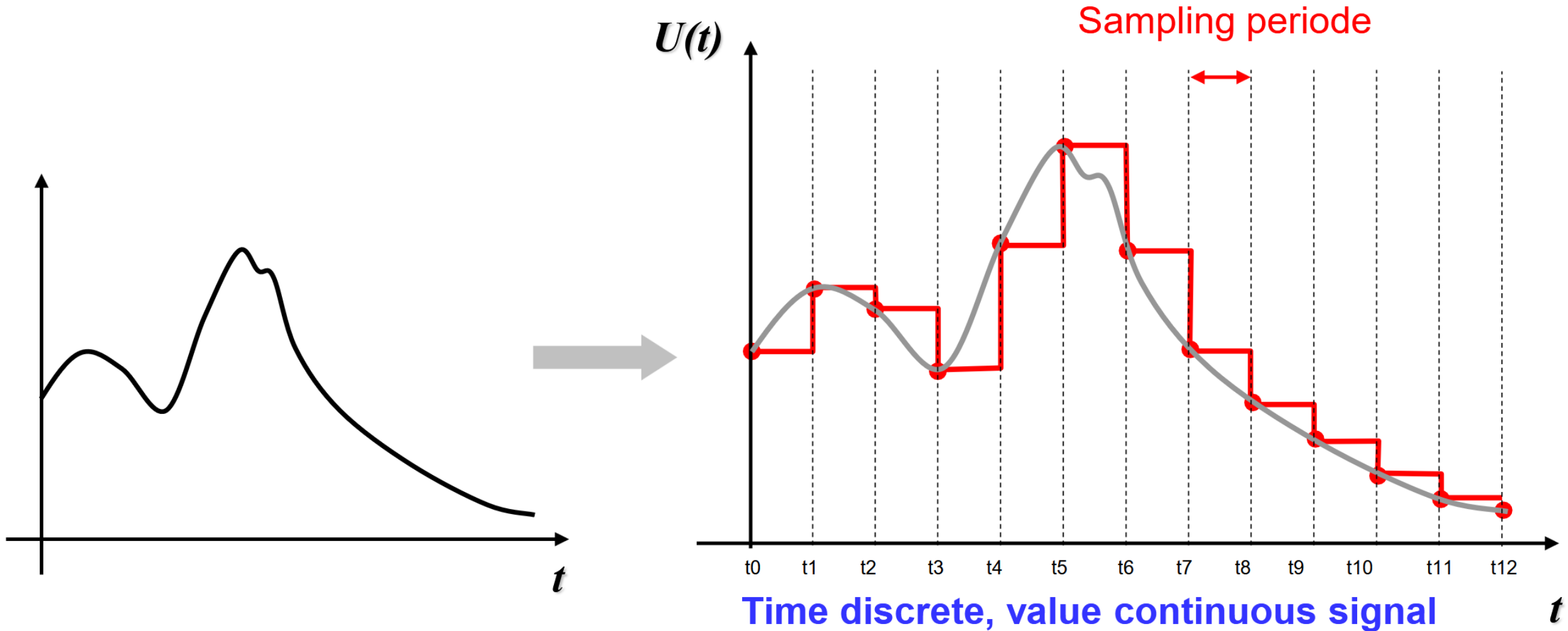
Analog-to-Digital Conversion

Nyquist–Shannon Sampling Theorem

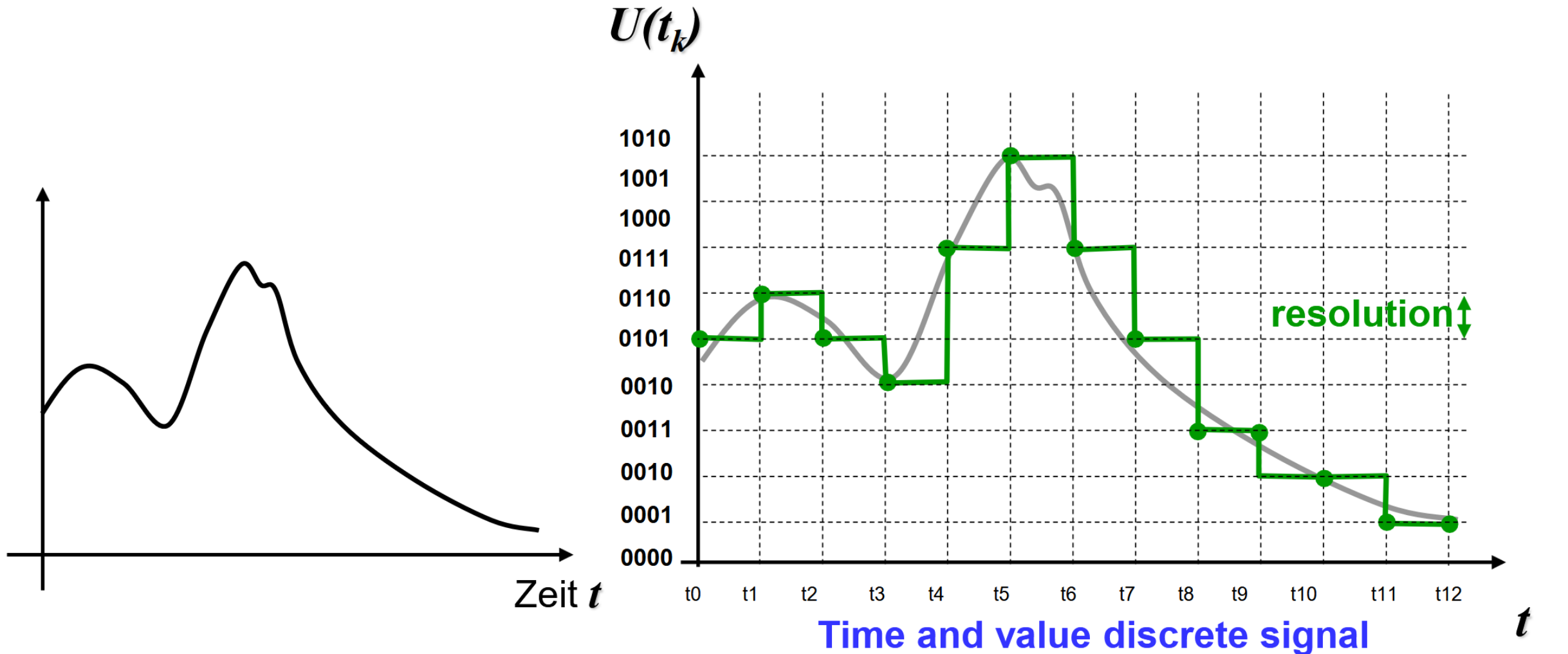
To perfectly reconstruct a (bandlimited) continuous signal from its samples, the sampling rate must be at least twice the highest frequency present in the signal.



Analog-to-Digital Conversion



Analog-to-Digital Conversion



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Classification of Sensors

Sensors can be classified based on

Output Signal

- Analog
- Digital

Energy Emission

- Active
- Passive

Measurement Principle

- Mechanics
- Optics
- Magnetics
- Acoustics
- ...

Measurement Target

- Proprioceptive (Internal)
- Exteroceptive (External)

Analog vs. Digital Sensors

- **Analog Sensors:**

- Produce (continuous) **analog output signals**
- Require analog-to-digital (A/D) conversion
- Examples:
 - Analog infrared distance sensor
 - Microphone
 - Analog compass

- **Digital Sensors:**

- Generate discrete **digital output signals**
- Outputs can be delivered via :
 - Synchronous serial: data transmitted bit by bit
 - Parallel: multiple digital output lines (e.g. 8 bit or or 16 bit)
- Examples:
 - Digital camera
 - GPS

Active vs. Passive Sensors

■ Active Sensors

- **Emit energy** into the environment to gather information
- Require energy for operation
- Measure the reflected or returned signal (the feedback) to interpret the environment
- Examples:
 - Infrared sensor
 - Laser range finders
- Characteristics: more robust, but less energy-efficient

■ Passive Sensors

- Monitor the environment without actively affecting it
- **Detect energy** already present in the environment
- Examples:
 - Vision camera
 - Gyroscopes
 - Temperature probes
- Characteristics: less intrusive, but dependent on environment conditions

Measurement Principle

Mechanical Principle

- Require **physical contact** between robot and sensor
- Often integrated directly into the robot body
- Provide tactile feedback and force measurements

Acoustic Principle

- Utilize ultrasound frequencies for **distance** measurement
- Measure **directionality** and **time-of-flight** of sent and received signals
- Calculate distance based on signal return time

Electromagnetic Principle

- Similar to acoustic systems using **directionality** and **time-of-flight** measurement
- Require clear "line of sight" between transmitter and receiver
- Examples include LIDAR and radar systems

Measurement Principle

Magnetic Principle

- Use **Earth's magnetic field** and **solenoids** for position calculation
- Determine orientation based on spatial configuration of magnetic fields
- Common in compass and navigation applications

Optical Principle

- Utilize vision **cameras** (monocular, binocular, omnidirectional)
- Provide **rich visual information** for environment perception
- Enable object recognition and spatial awareness

Measurement Target

Proprioceptive (Internal) sensors

- Measure of the internal state of the robot
- Examples: Joint angle encoders, torque sensors, IMU

Exteroceptive (External) sensors

- Measure the state of the environment
- Examples: Cameras, laser sensors

Sensor Taxonomy

Elementary Sensors

Record raw measured values or image signal and provide unprocessed data

Exp.: Photodiode, CCD sensor

Integrated Sensors

Include signal processing: amplification, filtering, linearization, normalization

Exp.: CMOS sensor

Intelligent Sensors

Integrated sensor with intelligent (real-time) processing using ML/AI methods and produce interpretation of the sensor data

Exp.: Digital camera with face-recognition

Trends show an evolution of intelligent sensors to **AI-enhanced** and **edge-computing** sensors!

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Sensors and Perception in Robotics: Task and Challenges

Task: Capture and interpret the state of the environment

Challenges

1. Partial information

- Sensors provide only partial view of the environment
- Challenge: Select appropriate sensors or sensor combination

2. Modeling of the sensor characteristics

- Determine the mapping between real-world phenomena and sensor measurements
- Challenge: Understanding sensor limitations and error characteristics

3. Digital Signal Processing

- Process raw sensor data to extract meaningful information
- Challenge: Applying signal processing techniques and machine vision algorithms

4. Multi-Sensor Integration

- Combine data from multiple sensors effectively
- Challenge: Sensor fusion for comprehensive understanding

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Sensors in Robotics – Examples

- Cameras
 - RGB
 - RGB-D
 - Stereo
- Joint angle encoders
 - Incremental (relative) encoders
 - Absolute encoders
- Inertial sensors
 - Accelerometers
 - Gyroscopes
 - IMUs
- Force/torques sensors
- ...



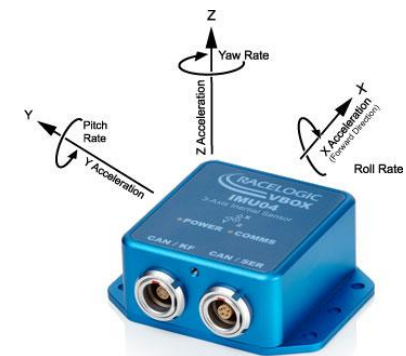
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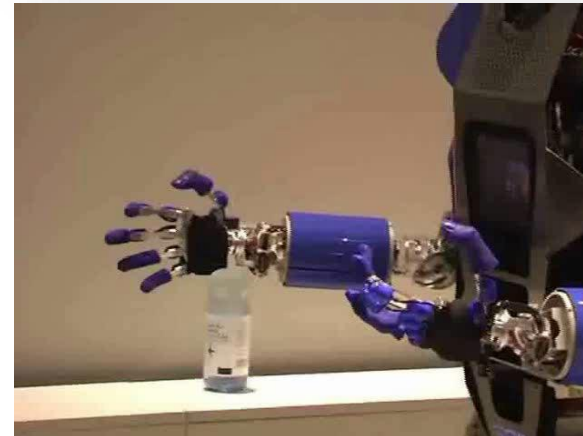
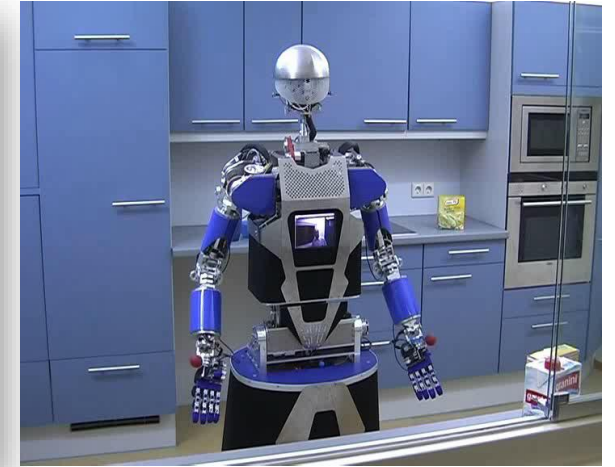
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Example: Sensors in ARMAR-III

- 7 DOF head with foveated vision
 - 2 cameras in each eye
 - 6 microphones
- 7-DOF arms
 - Position, velocity and torque sensors
 - 6D FT-sensors
 - Sensitive skin
- 8-DOF hands
 - Finger position sensors
 - Tactile sensors
- Holonomic mobile platform
 - 3 laser scanners



Example: Sensors in ARMAR-6

- Sensor-Actuator-Controller Units in each arm joint
 - Incremental position (motor)
 - Absolute position (output)
 - Torque (output)
 - Motor current
 - IMU
 - Temperature sensors (motor, motor controller, gear)
- Head
 - RGB-D (Azure Kinect)
 - 2 RGB cameras (Point Grey Flea)
 - Stereo-sensor (Roboception rc_visard)
- Mobile base
 - Absolute position torso (draw wire sensor)
 - 2 laser scanners
 - Wheel positions (incremental)



Thank you

Next Chapter: **Proprioceptive (Internal) sensors**